

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**1<sup>st</sup> Semester of M. Pharm (Pharmaceutical Technology)**  
**University Theory Examination December 2015**  
**PH819.03 NOVEL DRUG DELIVERY SYSTEMS - I**

**Date: 21/12/2015, Monday**

**Time: 10.00 a.m to 1.00 p.m**

**Maximum Marks: 70**

**Instructions:**

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

**SECTION-I**

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|------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.1</b> | <b>A)</b> | Write a note on Drug Release Mechanism for Oral controlled drug delivery system.                                                      | <b>05</b> |
|            | <b>B)</b> | Explain various approaches of GRDDS with diagram and factors controlling it.                                                          | <b>05</b> |
| <b>Q.2</b> |           | Enumerate the factors affecting mucoadhesion. Describe formulation approach and evaluation test of mucoadhesive drug delivery system. | <b>10</b> |

**OR**

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|------------|------------|---------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.2</b> | <b>A)</b>  | Enumerate four drugs available as implants and inserts? Explain the evaluation of implants and inserts. | <b>05</b> |
|            | <b>B)</b>  | Nanohydrogels: As a novel drug delivery technique.                                                      | <b>05</b> |
| <b>Q.3</b> |            | <b>Answer the following (Any three)</b>                                                                 | <b>15</b> |
|            | <b>(A)</b> | Explain Osmotic Pump with labeled diagram.                                                              |           |
|            | <b>(B)</b> | Explain types of permeation enhancer used in TDDS based on their mechanism.                             |           |
|            | <b>(C)</b> | Explain various formulation components for fabrication of TDDS.                                         |           |
|            | <b>(D)</b> | Write a short note on Intra ocular implants.                                                            |           |

## **SECTION-II**

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|------------|----|------------------------------------------------------------------------------------|-----------|
| <b>Q.4</b> | A) | Describe the evaluation parameters of Fast Dissolving tablets                      | <b>05</b> |
|            | B) | Enlist patented technology for fast dissolving tablets. Explain any two in detail. | <b>05</b> |
| <b>Q.5</b> | A) | Differentiate between Emulsion and Microemulsion.                                  | <b>05</b> |
|            | B) | Explain any one theory of Microemulsion.                                           | <b>05</b> |

**OR**

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|------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.5</b> | A)  | Explain the mechanism of effervescent drug delivery system for controlled release of drug with examples of two marketed products.                                             | <b>05</b> |
|            | B)  | Explain Buccoadhesive film.                                                                                                                                                   | <b>05</b> |
| <b>Q.6</b> |     | <b>Answer the following (Any three)</b>                                                                                                                                       | <b>15</b> |
|            | (A) | Describe evaluation parameters such as Particle size, Conductivity, Zeta potential, Transparency and Entrapment efficiency for Microemulsion in detail with its significance. |           |
|            | (B) | Define Liposomes. Enlist various methods of preparation of liposomes. Explain in detail any one method with diagram.                                                          |           |
|            | (C) | Write a shortnote on: Application of Niosomes in Pharmaceutical field.                                                                                                        |           |
|            | (D) | Write a shortnote on: Microparticles as drug delivery system.                                                                                                                 |           |
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